CV5600
2026

Wildfire Prediction

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Introduction - Why California Matters

Global Fire Crisis

California faces frequent and destructive wildfires, with annual burned acreage increasing dramatically over the past two decades.

Climate Amplification

Rising temperatures extend fire seasons by months, while mega-droughts reduce soil moisture and increase fuel flammability.

Fuel Accumulation

Dense vegetation and dead biomass from bark beetle infestations create ideal conditions for intense, fast-moving fires.

Extreme Wind Events

Santa Ana and Diablo winds (50+ mph) create fire weather conditions that enable rapid spread across 10,000+ acres in a single day.

Study Area & Spatial Configuration

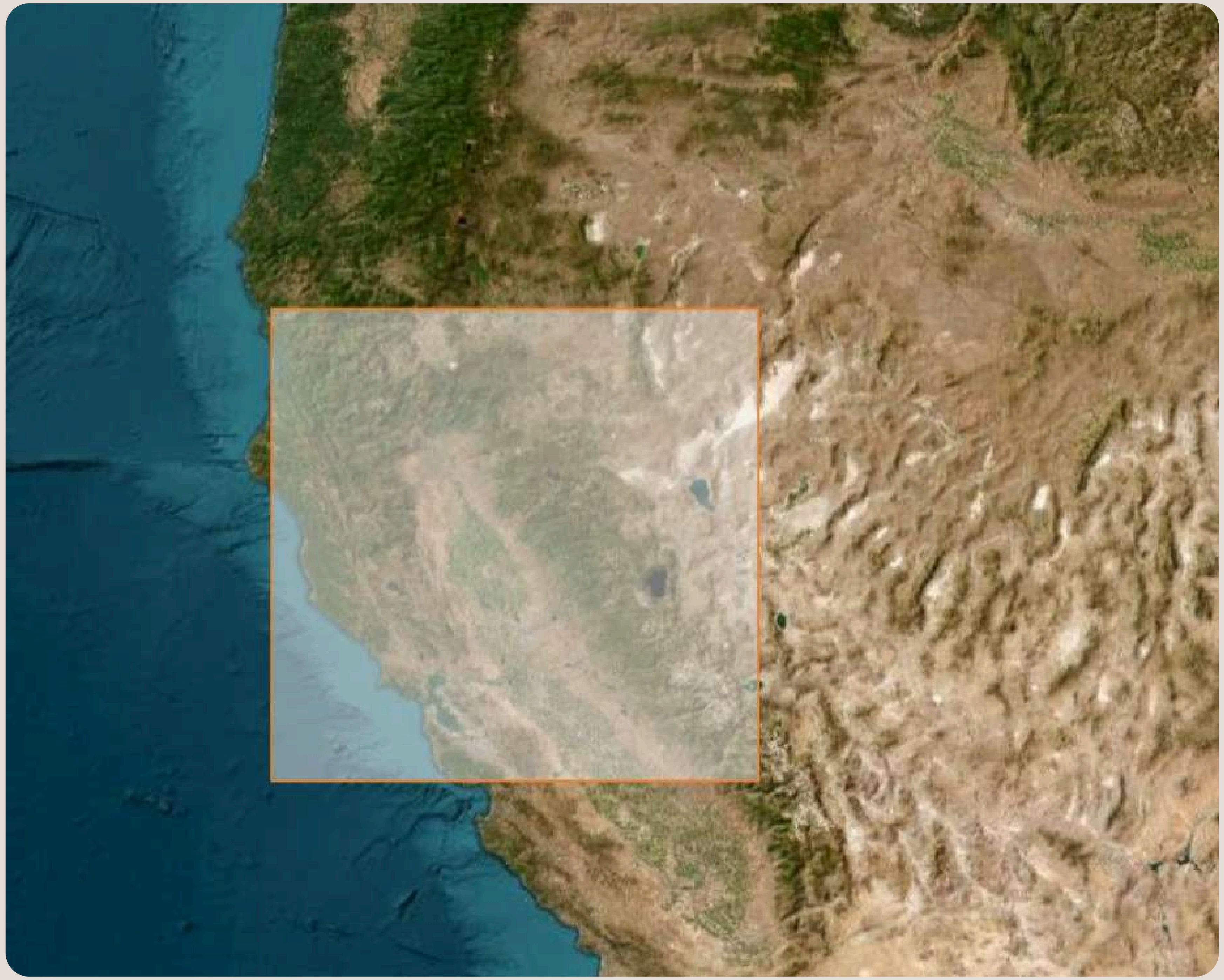
The study region is specifically selected to cover high-risk zones.

Bounding Box

Encoded as a GeoJSON poly-gon with corners at lon ∈ [−124.135, −118.963] and lat ∈ [36.993, 42.010], spanning approximately 580 km (east–west) × 560 km (north–south).

Temporal Scope

Analysis focuses on VIIRS S-NPP detections from 2012 to 2023.



Covers the Sacramento region in the north and Stanislaus National Forest in the south.

Data Collection : Labels

VIIRS

Data is obtained from NASA FIRMS, we have used archive download to get the bulk fire detections aat once.

Feature definitions:

- **frp (Fire Radiative Power):** Measures the pixel-integrated fire radiative power in megawatts (MW).
- **confidence:** Indicates the reliability of the fire detection. This is a critical filter during data preprocessing to eliminate false positives from the training set. Values include:
  - l - Low confidence, n - Nominal confidence, h - High confidence
- **type:** The inferred classification of the hot spot. This label allows you to isolate actual wildfires from industrial anomalies:
  - 0: Presumed vegetation fire (wildfire), 1: Active volcano, 2: Other static land sources (e.g., industrial smoke stacks), 3: Offshore detection

latitude	longitude	brightness	scan	track	confidence	bright_t31	frp	daynight	type
38.01915	-122.23704	303.6	0.38	0.36	n	281.64	0.7	N	2
41.48973	-122.52775	299.63	0.39	0.36	n	277.25	0.78	N	0
39.62056	-119.26095	315.07	0.42	0.38	n	280.42	1.87	N	2
39.94645	-122.23311	304.87	0.38	0.36	n	279.65	1.02	N	0
39.94577	-122.22865	313.67	0.38	0.36	n	280.46	1.03	N	0

Data Collection : Temporal Features

GRIDMET

Meteorological forcing is sourced from the University of Idaho GridMET product, accessed via the **THREDDS OPeNDAP** endpoint at thredds.northwestknowledge.net. GridMET operates at 124th degree resolution (**approximately 4 km × 4 km**), making it one of the finest-resolution **daily** meteorological gridded products available for the US.

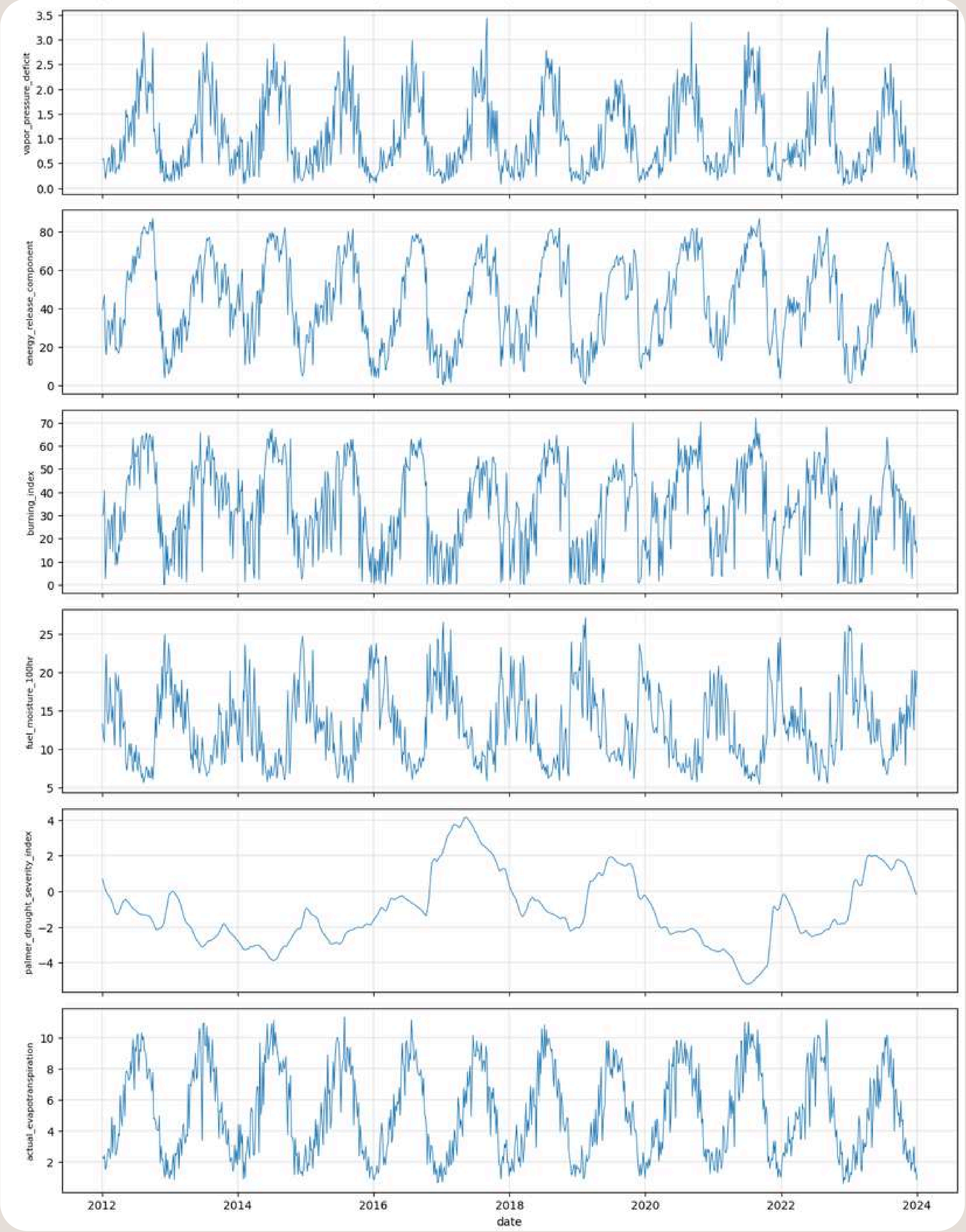
Each of the 17 variables is stored as a separate **NetCDF** file covering the full CONUS domain and is spatially subsetted to the bbox at download time. Downloads are parallelised using aiohttp and asyncio with up to four concurrent connections. Each file is opened lazily with xarray, merged across all variables, and streamed in 30-day chunks to control memory usage.

precipitation	max_temp	min_temp	wind_speed	solar_radiation	rh_max	rh_min	specific_humidity	energy_release_component	burning_index	fuel_moisture_100hr	palmer_drought_severity_index
0	290.600006	279.899994	7	131.300003	87.099998	41.700001	0.00556	30	39	16.799999	2.98
0	291.600006	279.5	6.4	130.600006	85.199997	37.299999	0.00538	31	39	16.1	2.41
0	292.200012	279.5	5.7	129.800003	83.199997	34.5	0.00529	34	44	15.5	2.21
0	292.600006	279.5	5.1	129.100006	80.5	33.5	0.00514	36	41	14.8	2.36
0	292.799988	280	4.5	127	76.300003	32.5	0.00505	39	41	13.8	2.4

Data Collection : Temporal Features

GRIDMET

Code	Column	Description
pr	precipitation	Daily total precipitation (mm)
tmmx	max_temp	Daily maximum temperature (K; converted to °C for FWI)
tmmn	min_temp	Daily minimum temperature (K)
vs	wind_speed	Daily mean wind speed (m/s; converted to km/h for FWI)
th	wind_direction	Daily mean wind direction (degrees from north)
srad	solar_radiation	Daily mean shortwave radiation at surface (W/m²)
rmax	rh_max	Daily maximum relative humidity (%)
rmin	rh_min	Daily minimum relative humidity (%)
sph	specific_humidity	Daily mean specific humidity (kg/kg)
vpd	vapor_pressure_deficit	Daily mean vapour pressure deficit (kPa)
erc	energy_release_component	Fire danger ERC-G index (dimensionless)
bi	burning_index	NFDRS Burning Index (dimensionless)
fm100	fuel_moisture_100hr	Dead fuel moisture, 100-hour lag (%)
fm1000	fuel_moisture_1000hr	Dead fuel moisture, 1000-hour lag (%)
pdsi	palmer_drought_severity_index	Palmer Drought Severity Index (dimensionless)
pet	potential_evapotranspiration	Reference ET, grass-based (mm/day)
etr	actual_evapotranspiration	Reference ET, alfalfa-based (mm/day)



Data Collection : Temporal Features

Spectral Bands

VIIRS Spectral Reflectance and Land Surface Temperature

Daily spectral reflectance and land surface temperature (LST) are obtained from two NASA VIIRS product families accessed via the Google Earth Engine (GEE) Python API.

- VNP09GA (NASA/VIIRS/002/VNP09GA): **Daily 500 m surface reflectance**, exported as monthly GeoTIFFs to Google Drive and downloaded via the Drive API. Monthly chunking avoids GEE's 1024-band-per-export ceiling (31 days × 6 bands = 186 bands per file).
- VNP21A1D / VNP21A1N (NASA/VIIRS/002/VNP21A1D, VNP21A1N): **Daily 1 km LST**, daytime and night-time acquisitions respectively.

Table 3: VIIRS spectral and thermal bands collected.

Band	Column	Description
I1	i1	Red, ∼600–670 nm; surface reflectance
I2	i2	NIR, ∼846–885 nm; primary burned-area discriminator
I3	i3	SWIR, ∼1.23 μm; vegetation/burn differentiation
M11	m11	SWIR, ∼2.225 μm; sensitive to fire and burn scars
NDVI	ndvi	$(I2 - I1)/(I2 + I1)$; computed on-the-fly in GEE
EVI	evi	$2.5(NIR - RED)/(NIR + 6 \cdot RED - 7.5 \cdot BLUE + 1)$
LST (day)	lst_day	Land Surface Temperature, daytime (K), 1 km
LST (night)	lst_night	Land Surface Temperature, night-time (K), 1 km

All GEE exports use crs="EPSG:4326" (WGS84). Each exported band is labelled YYYYMMDD <BandName>, allowing the transform script to reconstruct the time axis by regex-parsing band descriptions. Nodata sentinels set by GEE are replaced with np.nan; cloud-covered, off-swath, and ocean pixels are thus excluded from training. Since VNP09GA operates at 500 m and LST products at 1 km, the merge is performed via snapping 500 m pixel centres to their nearest 1 km LST pixel.

Data Collection : Static Features

DEM & Landcover

SRTM Terrain — Elevation, Slope, and Aspect

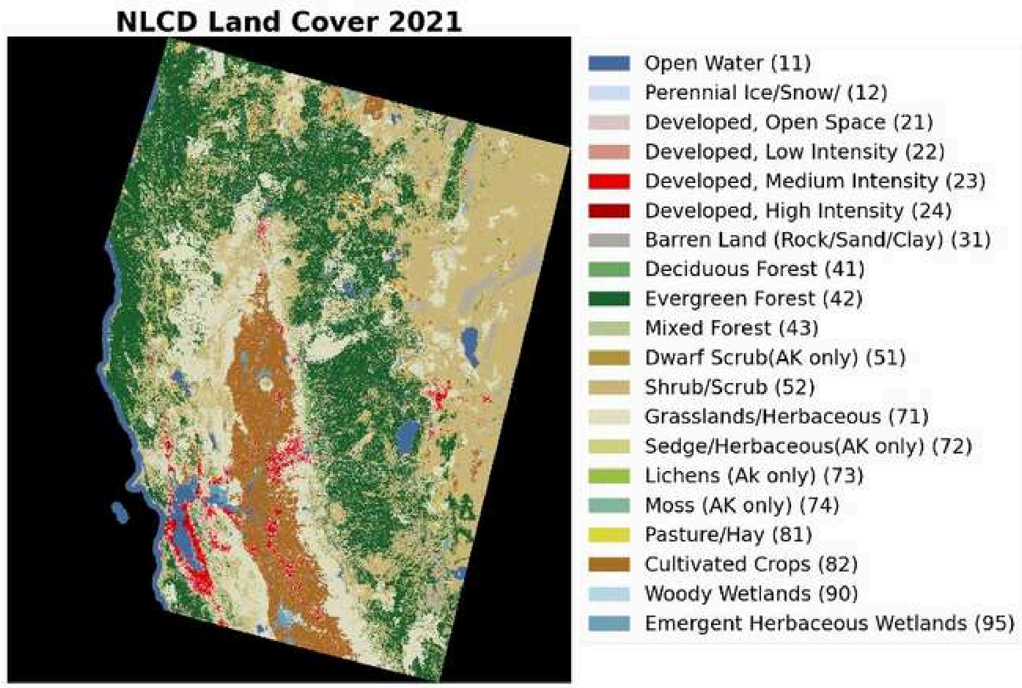
Static terrain variables are derived from the CGIAR Shuttle Radar Topography Mission (SRTM) - digital elevation model (DEM), accessed via the Python elevation package at **30 m native resolution**, clipped to the California bounding box. The raw GeoTIFF is read using rasterio; the no data sentinel value (typically −32768) is replaced with np.nan, and pixel-centre geographic coordinates are recovered via the affine transform.

Table 4: NLCD land cover class mapping to Anderson-13 fuel types.

NLCD Code	Label	Anderson Fuel Type
11	Open Water	98
41	Deciduous Forest	8
42	Evergreen Forest	9
43	Mixed Forest	10
52	Shrub/Scrub	6
71	Grassland/Herbaceous	3
82	Cultivated Crops	4
90	Woody Wetlands	11
21–24	Developed (various)	99

Land Cover — NLCD

Static land cover is sourced from the Multi-Resolution Land Characteristics Consortium (**MRLC**) National Land Cover Database (NLCD) at **30 m resolution** in Albers Equal Area Conic projection (EPSG:5070). The dataset is reprojected to WGS84 as described in Section 3.2. Each pixel's NLCD class code is mapped to an Anderson-13 fuel type category to provide a fire-behaviour-relevant encoding of surface cover.



Pipelines Implemented

wmWws : we model What we see

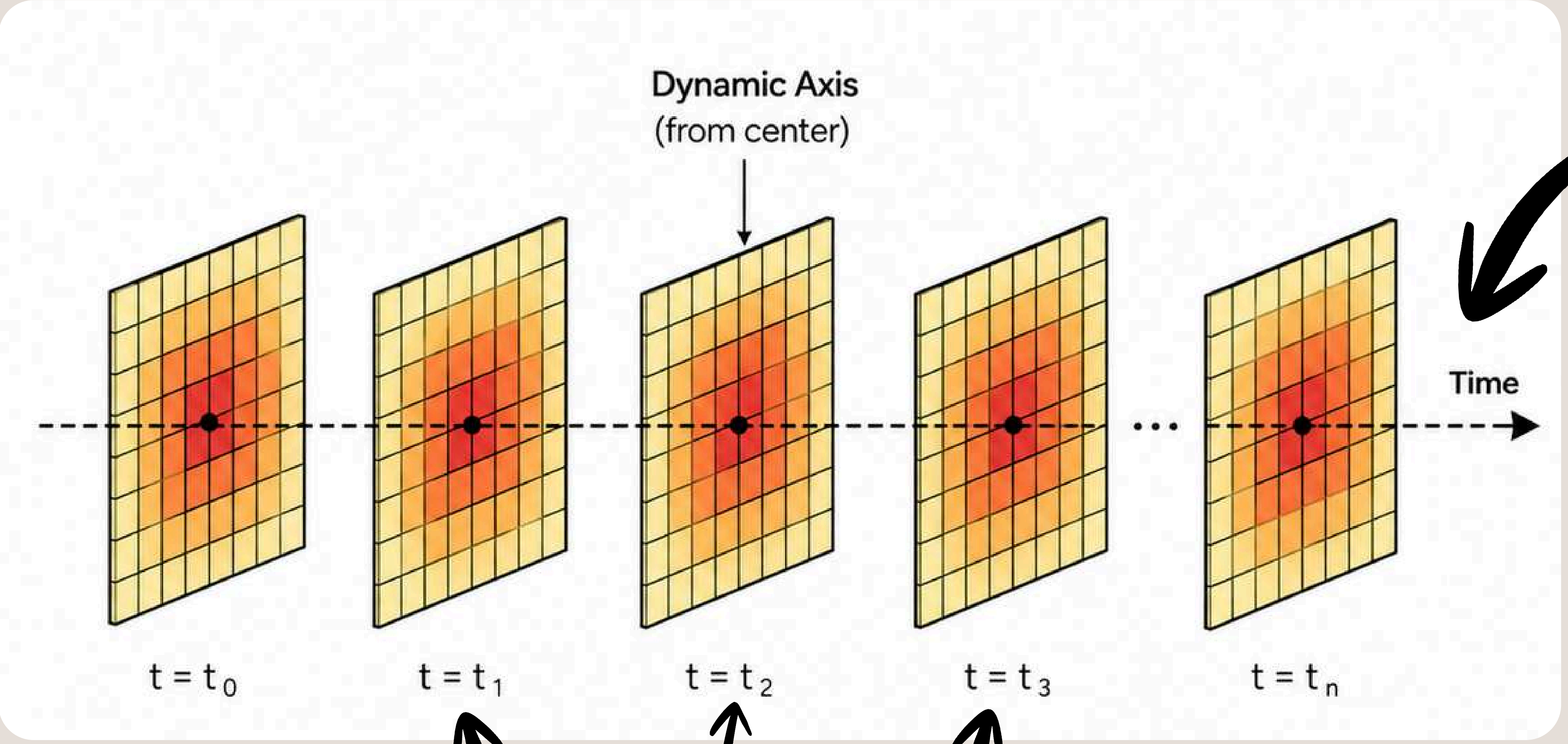
- Learns directly from **raw** spatial and temporal wildfire observations.
- Pipeline mimics a natural flow about how a layman person would **brute-force ML**
- Off-course a naive ideology which is bound to make mistakes.

wmWwwTs : we model What we want To see

- Transformed the dataset to capture the **required variance** from raw dataset
- Made mathematical **assumptions** to simplify the real world dynamics.
- Improved correctness by modeling the behavior closer to real dynamics.

wmWws : we model What we see

- **Unit of learning:** One full day = one full wildfire map
- **Spatial frame:** Fixed study-area grid (40 x 42, 4 km cells), same layout every day.
- **Input tensor per sample:** Historical daily stack X ∈ [Temporal, Channel, Height, Width]
- **Label per sample:** Same-day binary fire map Y ∈ [H, W] (fire / no-fire per cell).



Spatial Data Capture

Temporal Data Capture

wmWws : we model What we see

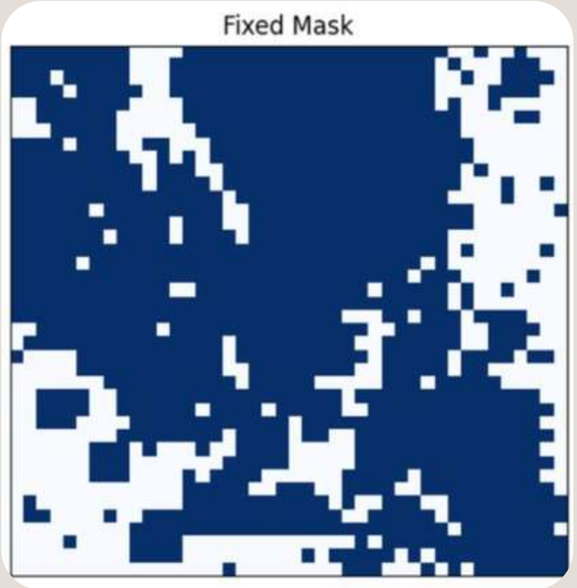
Label Generation

Phase 1: Raw Fire Image Generation

- Converted fire-point detections into daily full-grid fire maps, **EPSG:4326 to EPSG:32610**
- Effective class imbalance was around **641+** negatives per positives.

Phase 2: Heuristic Fire Signal Cleaning

- Sweep many candidate rectangular AOIs (grid windows, bounded size like **<= 60x60**)
- AOI fire density optimization by **shifting, expanding and contracting** grid heuristically (**194** Nofire:Fire)
- Score based Mask optimization for high-bounty regions by **top-k fire-frequency ranking + area sweep**, **hotspots + dilation**, **density-weighted sampling**, **dynamic programming - Knapsack**
- Optimized the score of masks based on 0.70*recall + 0.20*precision + 0.10*context - 0.05*area.
- Final effective **Fire : No Fire** ratio was **1 : 140**



Static Mask covering 70% of entire Area

wmWws : we model What we see

Feature Generation

Weather and derived data:

- **Daily gridMET variables** (2012–2023) extracted from standard .nc (NetCDF) formats.
- **Atmospheric Features:** Includes temperature extremes (tmmx/tmmn), wind_speed, humidity (rh_max/min), vapor pressure deficit (vpd), and precipitation.
- **Fire Behavior Indices:** Integration of the Canadian FWI System components: Fine Fuel (FFMC), Duff (DMC), Drought Code (DC), Initial Spread (ISI), and Buildup (BUI).

Vegetation health indices:

- Obtained from MODIS API, **16 day & 1km resolution**
- **Output:** NDVI, EVI, Pixel-reliability
- Cloud and snow pixels are NaN'd out to reduce data corruption

Spatial Preprocessing

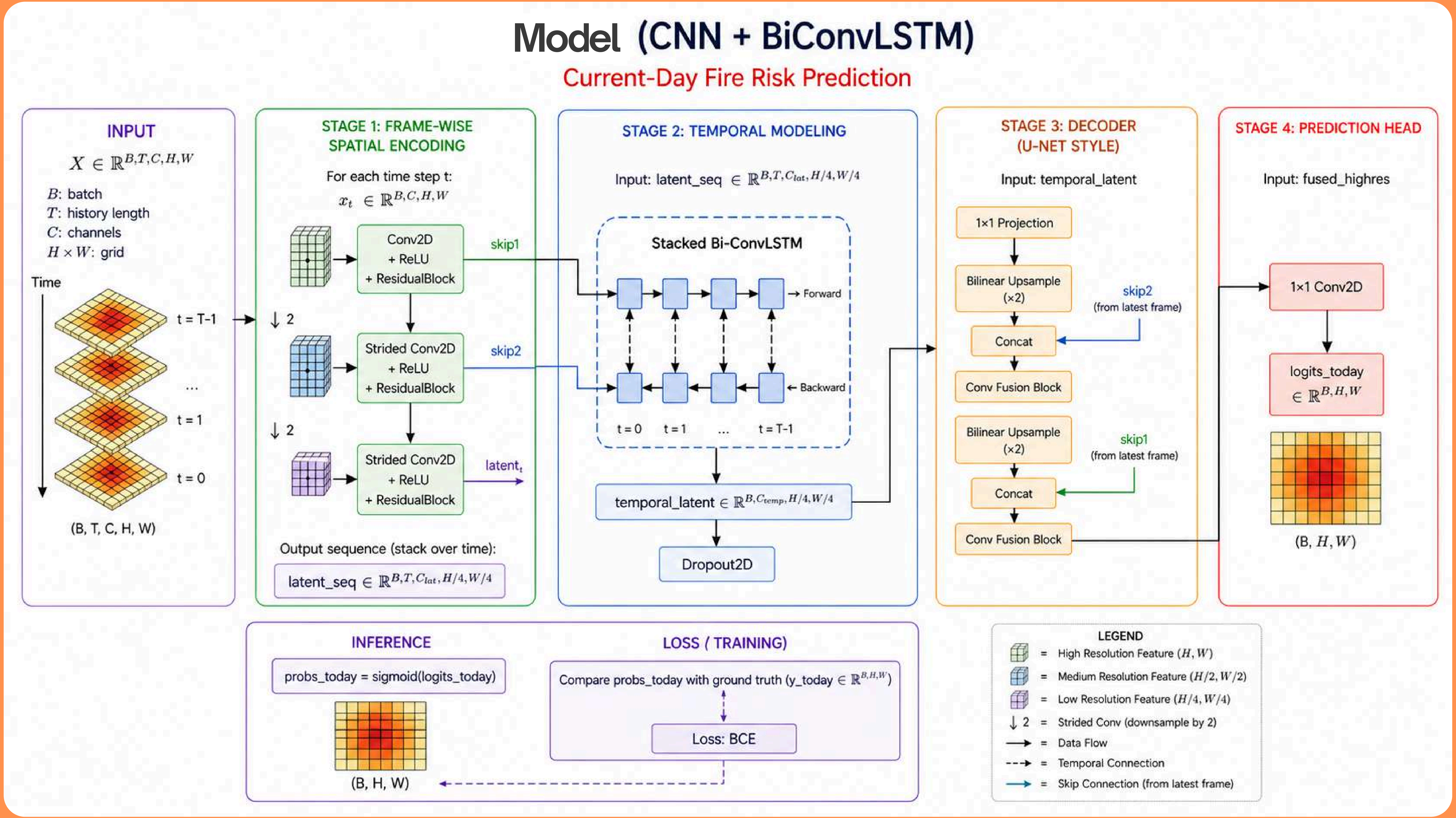
- Automated latitude normalization and cubic spline resizing using scipy.ndimage.zoom to reach the **target [40, 42]** grid shape.

```
feature_map shape = [C,H,W] = [20,40,42]

Channel index -> feature name
00 -> bui
01 -> dc
02 -> dmc
03 -> erc
04 -> evi
05 -> ffmc
06 -> fwi
07 -> isi
08 -> max_brightness
09 -> max_frp
10 -> n_detections
11 -> ndvi
12 -> precip
13 -> rh_max
14 -> rh_min
15 -> sum_frp
16 -> temp_max
17 -> temp_min
18 -> vpd
19 -> wind_speed
```

Final feature stack

wmWws : we model What we see

Model Arch



Captures Temporal and Spatial patterns

wmWws : we model What we see

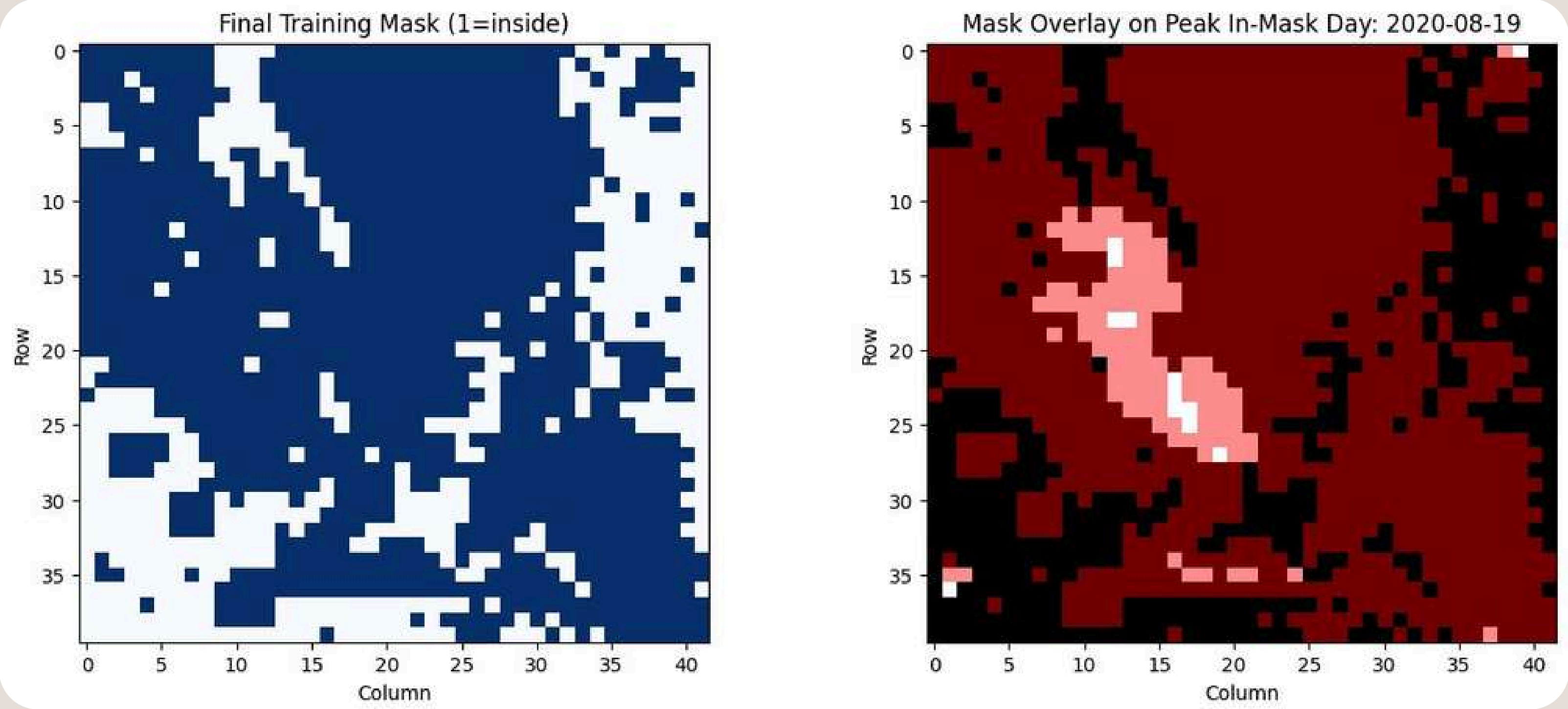
Problems

Identifying** what **may be wrong with the model???

Model is strong enough to capture the patterns, Problems may arise if the data quality is bad

Analysing the data quality before concluding the model capacity

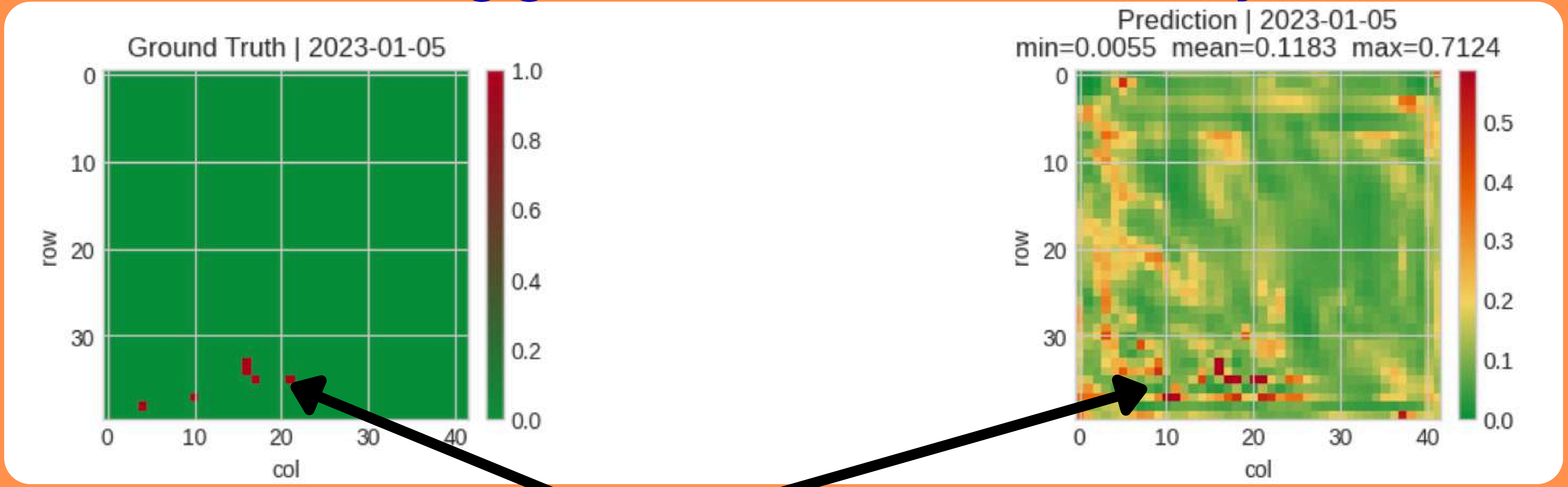
- **R**eal signal maybe **sparse** + **noisy** + off-course highly imbalanced
- **M**any samples may-be easy background, few were informative fire **transitions**
- **Temporal patterns maybe** weak/inconsistent **at raw daily resolution.**
- **Optimization** would always **favored safer background** predictions



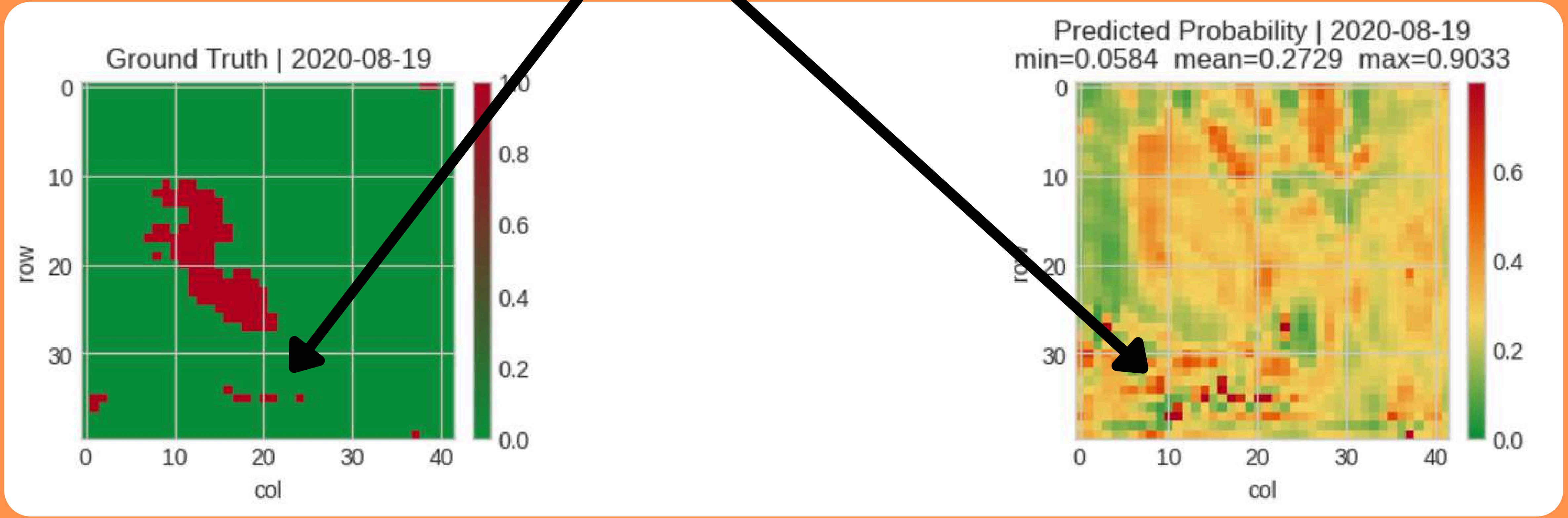
wmWws : we model What we see

Results

We may think the model is working good until we realize this maybe the case for each day



We can see that it has learnt the **temporal mean**, but exact segmentation is just gaussian.



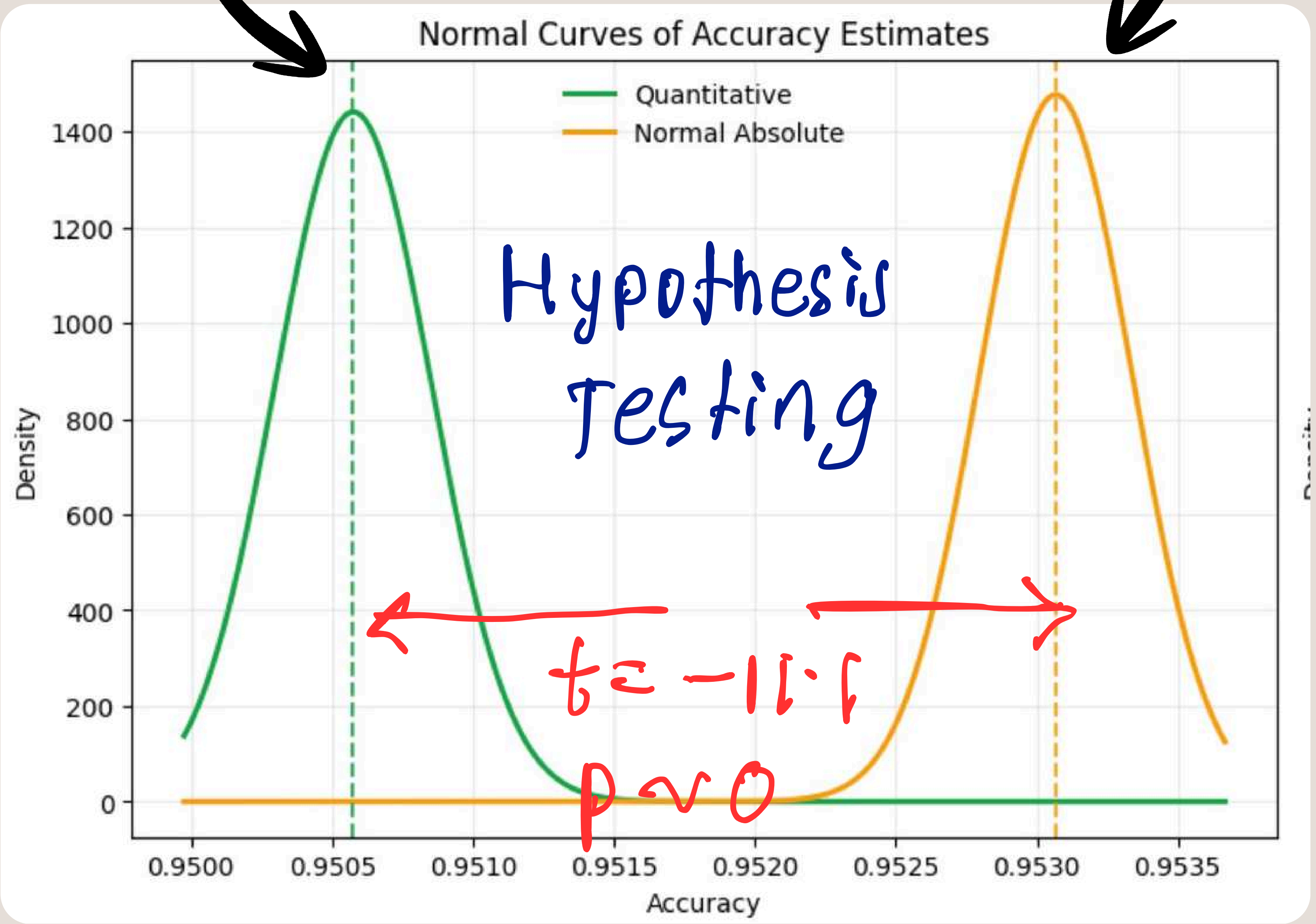
wmWws : we model What we see

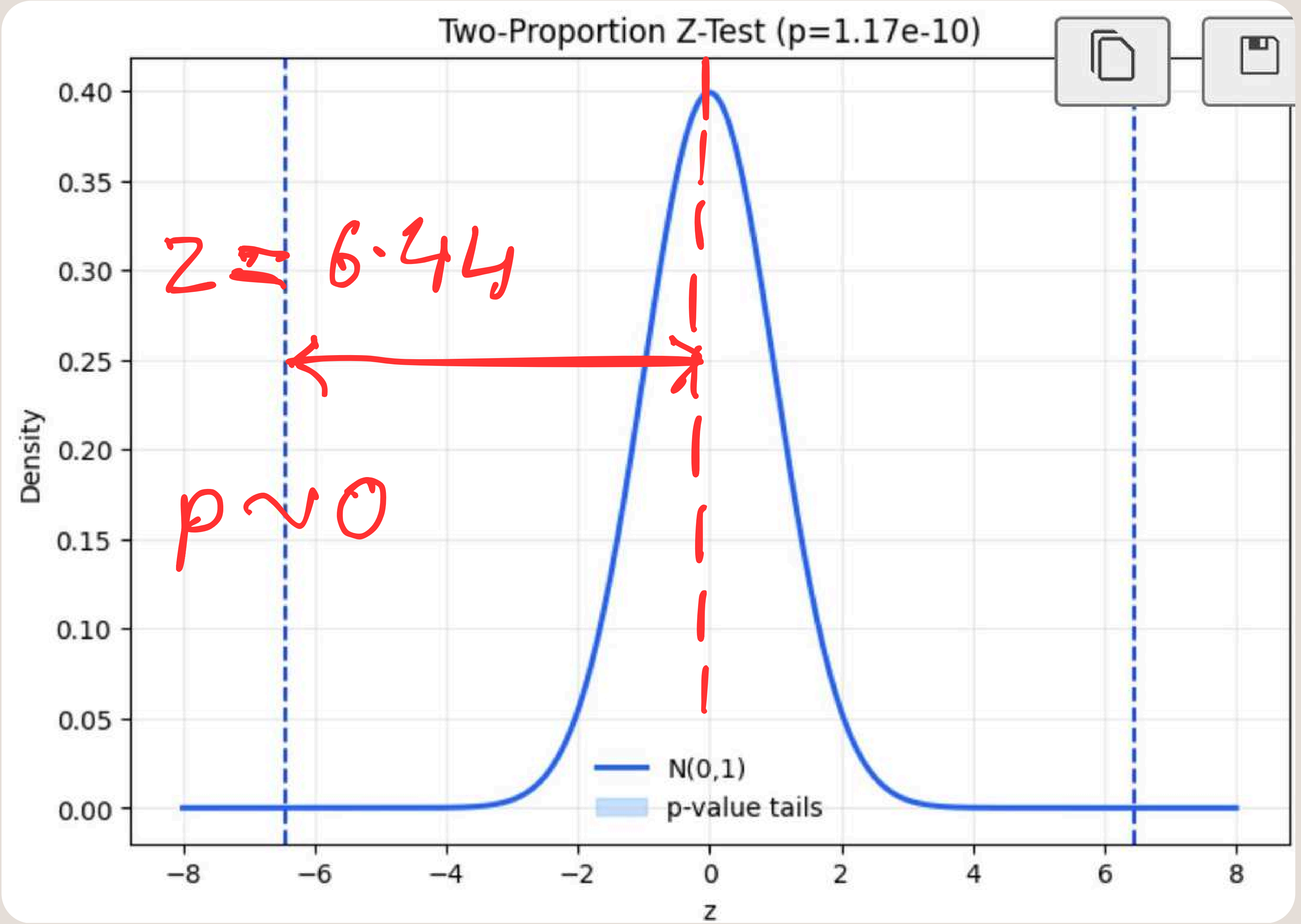
Validating the temporal layers

Multi-Collinear
Temporal features
Base Model ~ 111 features

16 - feature Model

Normal model was sufficient
so including the temporal
features made it worse





Terminal Stats Explained

```
Quantitative -> using label='y_true', prob='prob', keys=['timestamp', 'row', 'col']
Normal Absolute -> using label='y_true', prob='prob', keys=['timestamp', 'row', 'col']
Merge keys: ['timestamp', 'row', 'col']
Aligned rows: 613200
Label agreement: 1.0
Accuracy Quantitative: 0.9505756686236139
Accuracy Absolute: 0.9530675146771037
Accuracy diff (Quantitative-Absolute): -0.0024918460534898434
P-test z: -6.4432247870546835 p: 1.1696132951044547e-10
Paired t-test basis: daily_f1 t: -11.099537494949347 p: 7.481210126046707e-25
```

zero

Significant difference (not a random event)

Conclusion : Temporal Layers in the model are not the issue for current results

wmWwwTs : we model What we want To see

Rise of New Pipeline

Understanding what we want and what can simplified????

Given a **region** and **history** data??? What is the **probability** of fire **Today**???

- **wmWws** was doing it? **Yes**
- Was that reliable model? **Yes, verified with hypothesis testing**
- Was the data quality reliable? **No**

From Previous Result slides:

We noticed that the model is giving more importance to the **spatial dependency.**

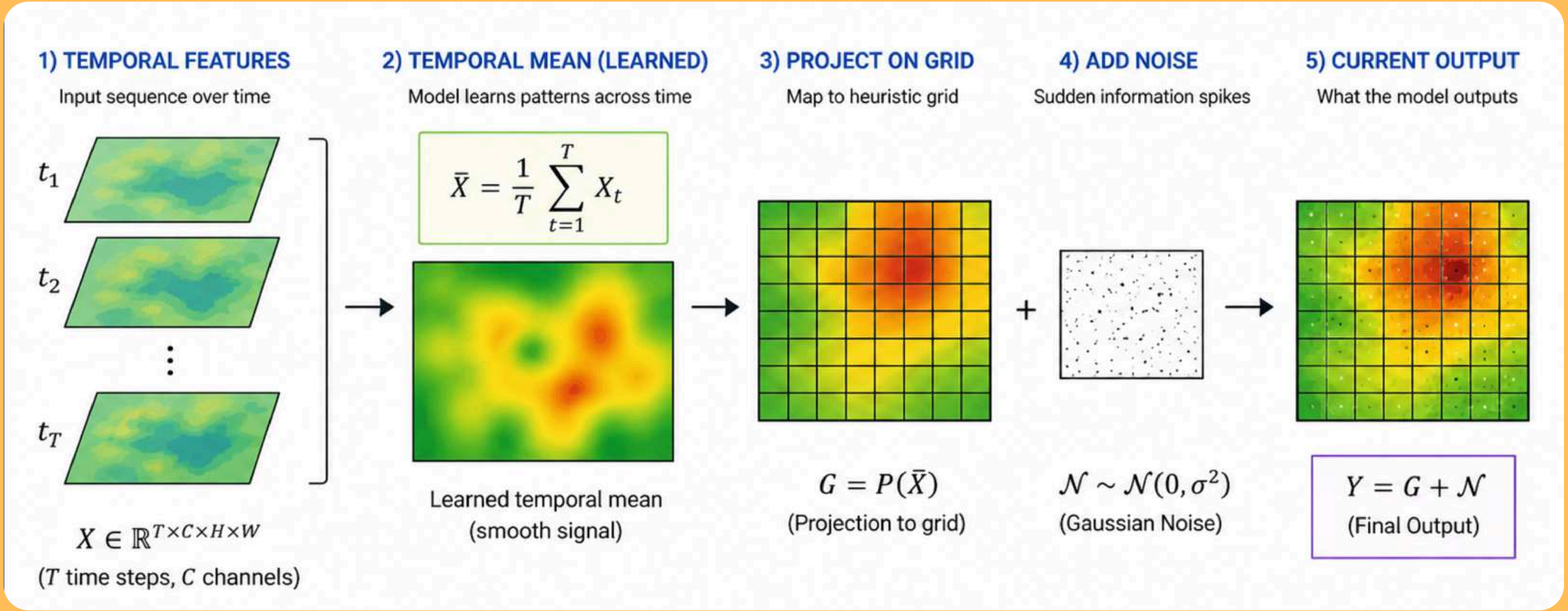
- Staring for a longer time at the plot, we noticed it was actually capturing the **temporal insights**
- It is the learning the **temporal gain** but it is influenced by **neighbouring** cells and effects are getting **smoothed by kernels.**

wmWwwTs : we model What we want To see

Rise of New Pipeline

The model learnt the **bounty cells** by temporal analysis, the bounties we unknowingly placed ourself while creating the the **mask** and the **heuristically defined grid.**

So the current output was **temporal mean** projected on the grid + **Gaussian Noise** by sudden **information spikes**



Since the temporal mean across the space wont change much in **15 days.** The model will return the same output each day with nearly the **same mean**

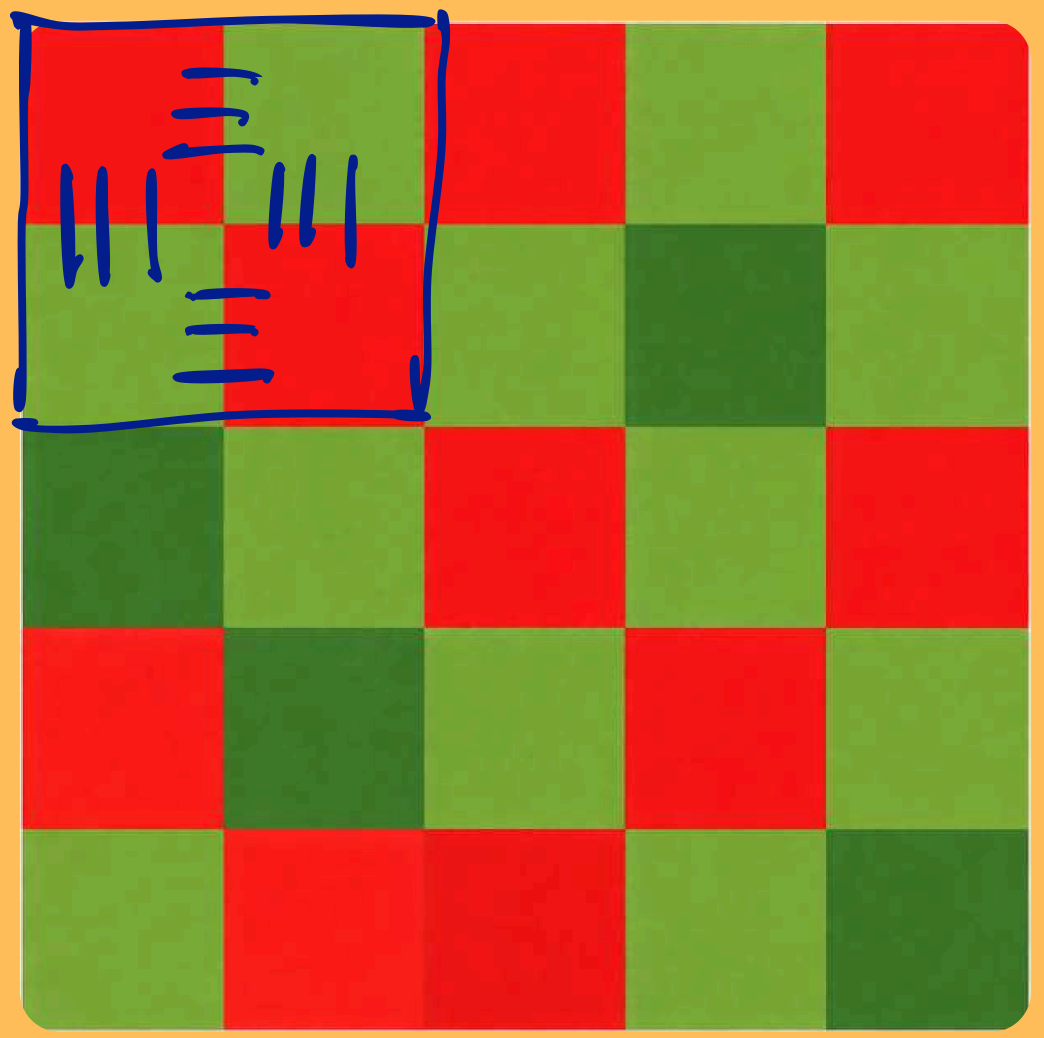
Solution: Collapsing the dimension

wmWwwTs : we model What we want To see

Assumptions

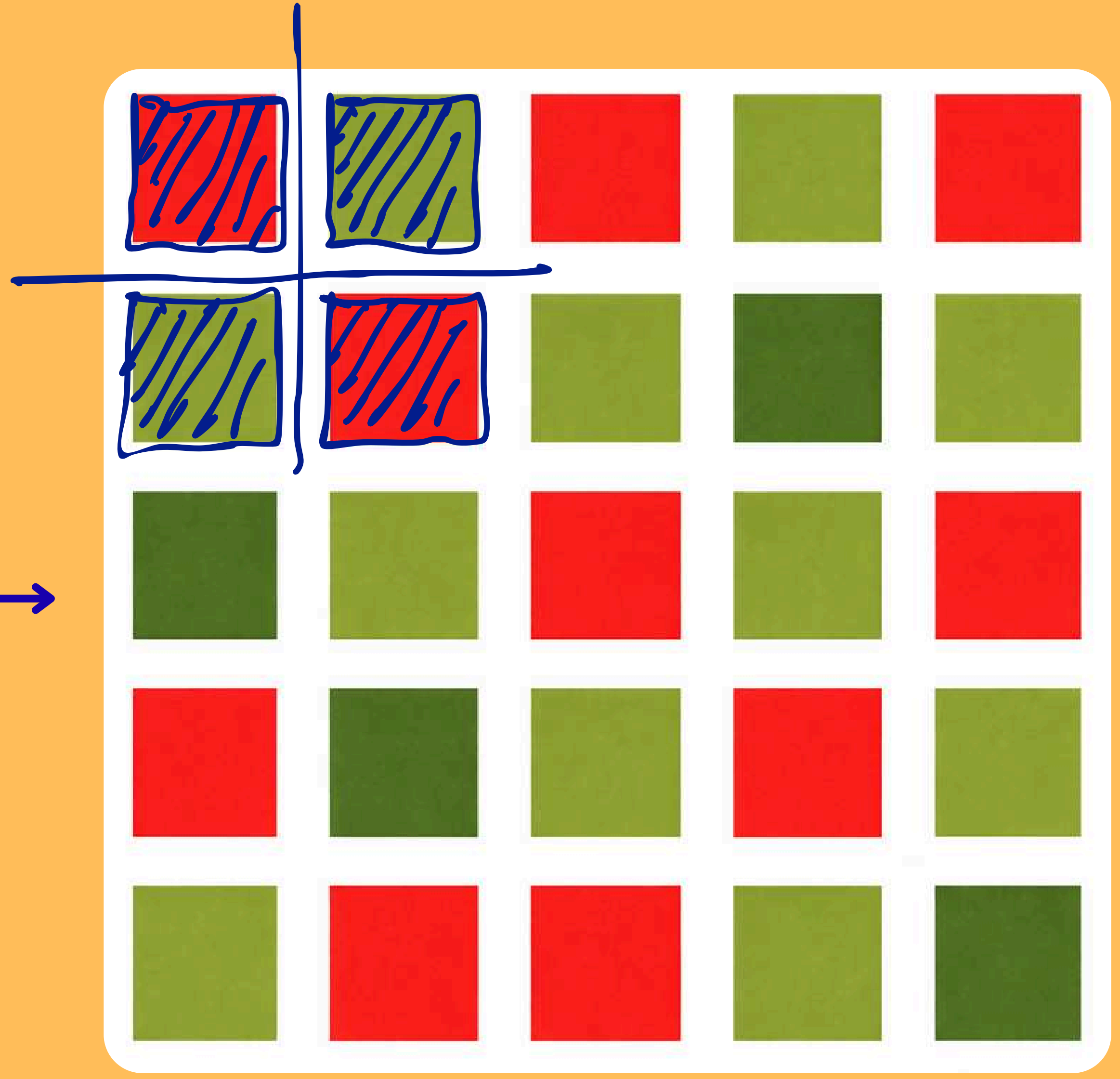
Spatial dependency captured by the kernel

Safe to assume 4 km x 4 km pixels
Area is independent to its neighbours



Eliminating

Dependency



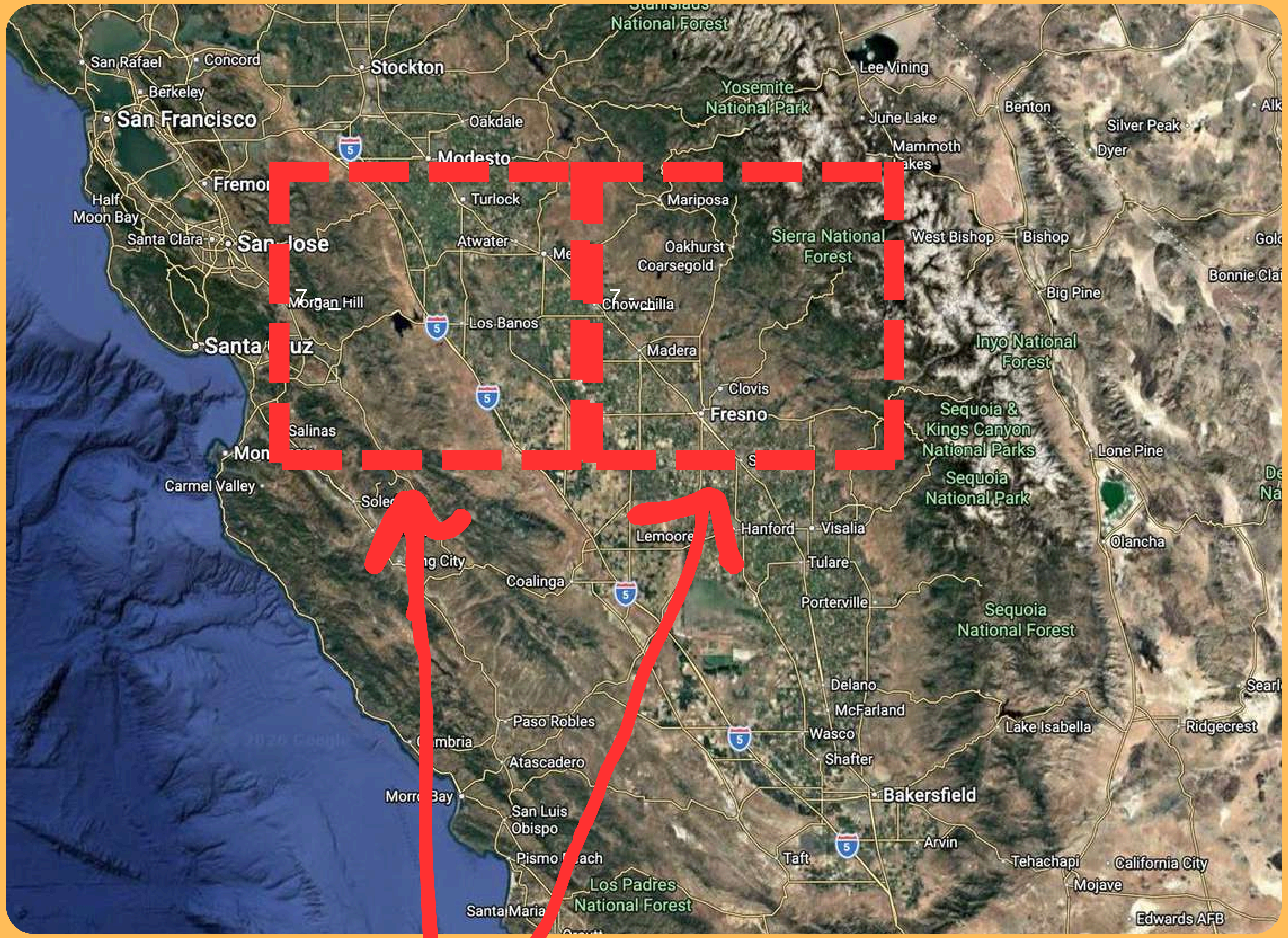
40 x 42 grid
represent individual sample

Each cell represents individual sample here

wmWwwTs : we model What we want To see

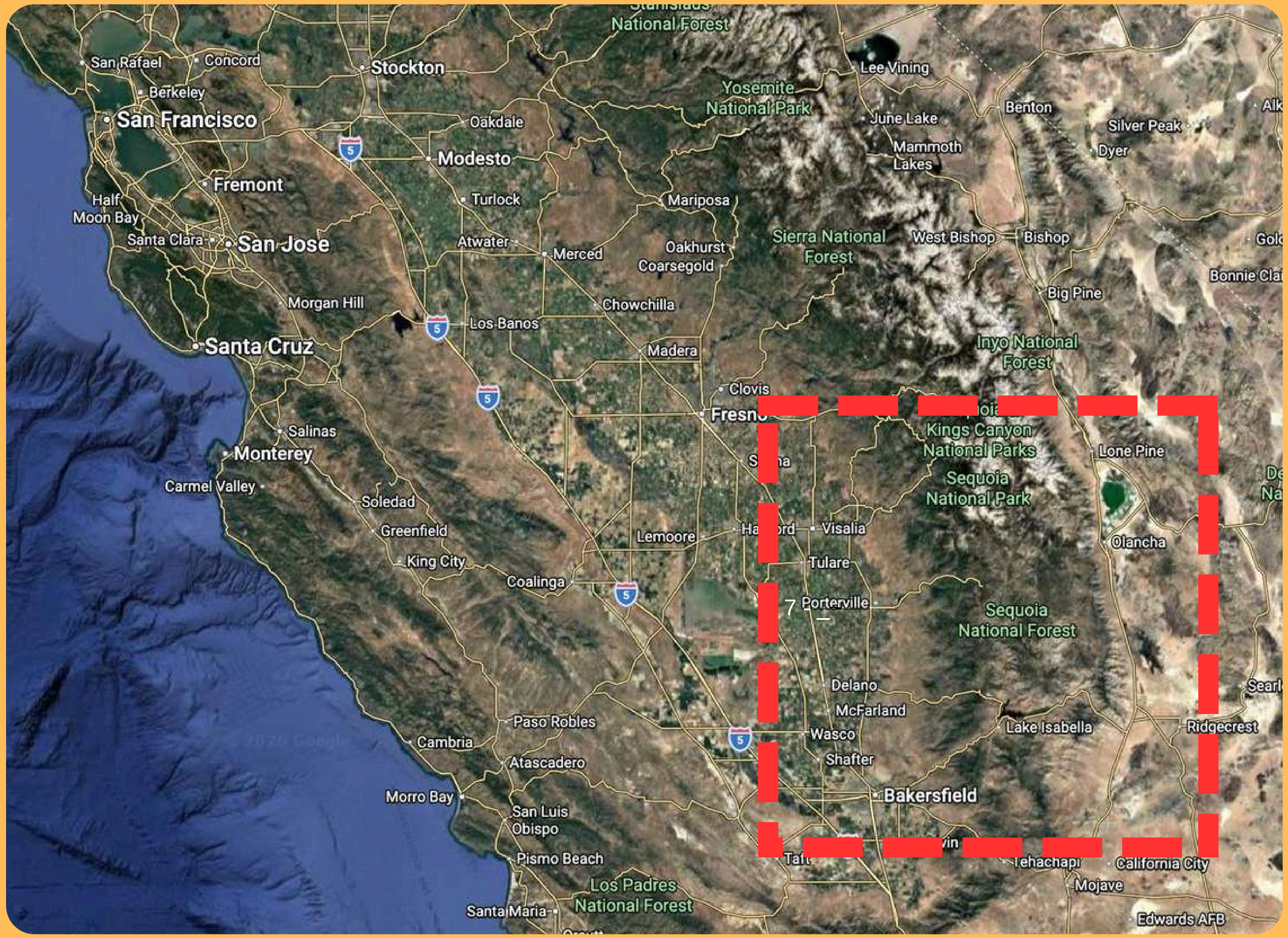
Benefits of Assumptions

California



Revised idea makes neighbour cells independent

Random Location



Allows us to to include any part of the globe in training data

wmWwwTs : we model What we want To see

Custom Dataset

We received the NDVI, EVI data in 16 days **temporal resolution (Limiting Feature)**

Historical 64 days split into (4) 16-day windows

For a **Window** to be labelled as 1, we expect atleast **3 fire points** in that particular window

We assume that if a particular cell had ignition point for more than 3 days in a window, Then it has enough FUEL to be a NATURAL FIRE irrespective of the fire been MAN-MADE or NATURAL

Fire Labels

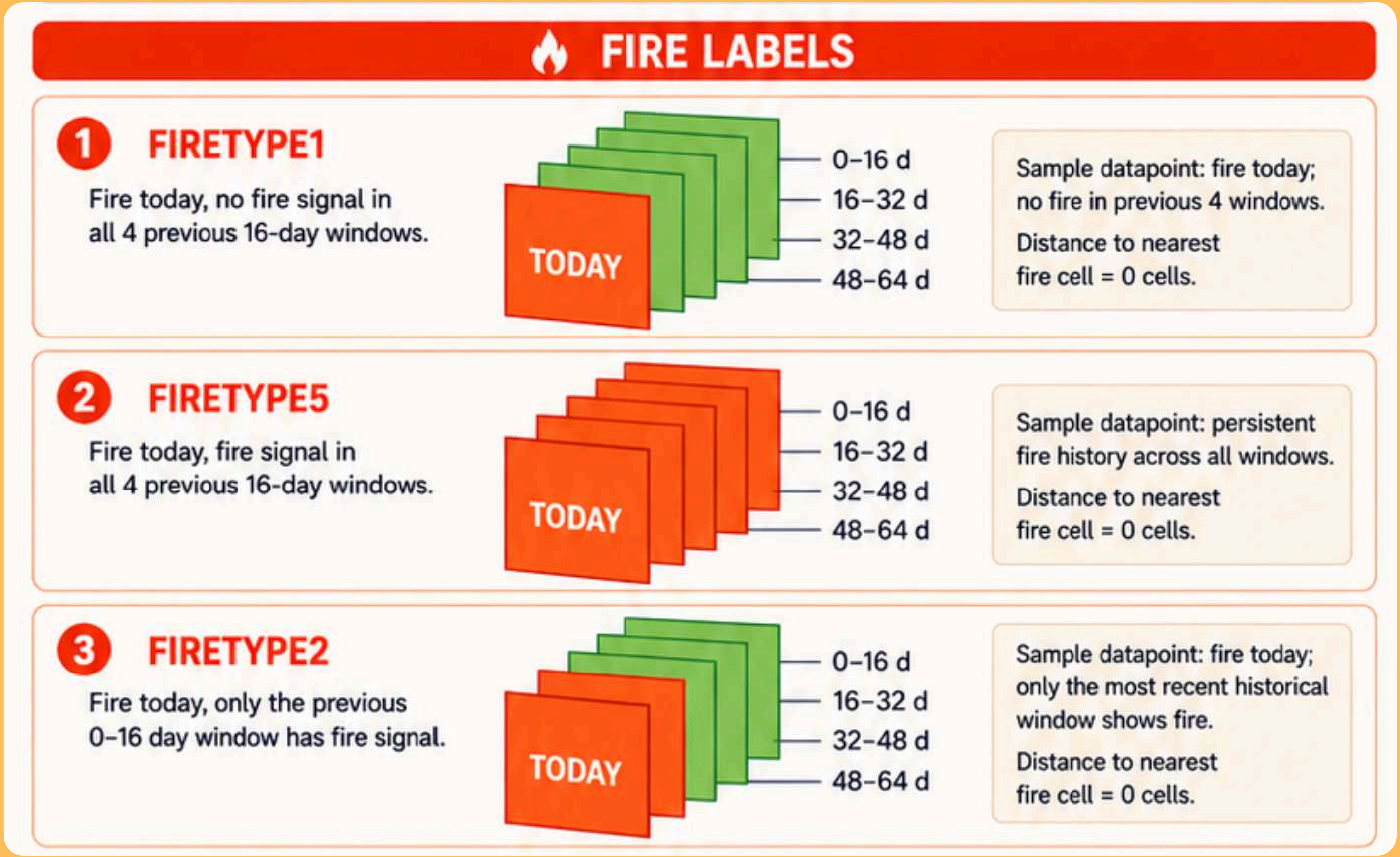
- **FIRETYPE1 :** Fire today, no fire signal in all 4 previous 16-day windows.
- **FIRETYPE5 :** Fire today, fire signal in all 4 previous windows.
- **FIRETYPE2 :** Fire today, only previous 0–16 day window has fire signal.

No Fire Labels

- **NOFIRE0A :** No fire today, distance to same-day fire in (1,5] cells, 0 in all windows
- **NOFIRE0B :** No fire today, distance (1,5] to fire cells, history fire sum >= 1 in 4 windows
- **NOFIRE0C :** No fire today, distance (5,10] to fire cells, history fire sum >= 1
- **NOFIRE0C :** No fire today, distance to fire cell >10 cells

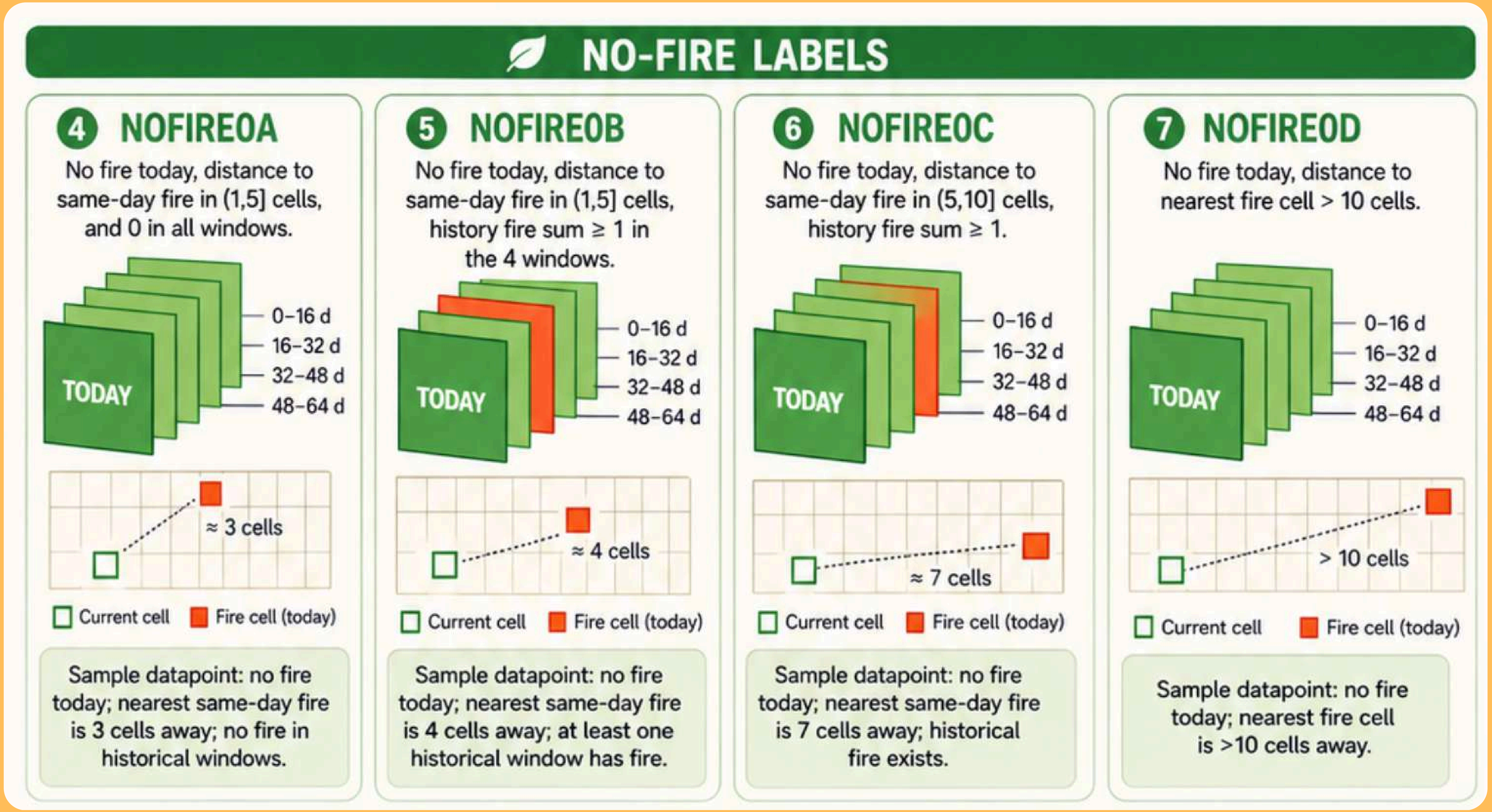
wmWwwTs : we model What we want To see

Fire Labels Example



wmWwwTs : we model What we want To see

No Fire Labels Example



This solves the major problem of class imbalance, while ensuring the model learns the temporal patterns in fire v/s no-fire (1:1)

wmWwwTs : we model What we want To see

X Feature Creation

For the chosen label datapoint, we constructed the **feature matrix**

▶ The full dataset viewer is not available (click to read why). Only showing a preview of the rows.

target_date string	row int64	col int64	window_id int64	aet float64	bui float64	burn_idx float64	dc float64	dmc float64	erc float64	ffmc float64
2012-04-01	7	46	1	0.525	1.097194	0	1.280599	1.168764	2.125	11.45528
2012-04-01	7	46	2	0.80625	2.553787	3.9375	3.048626	2.64127	6.0625	21.950106
2012-04-01	7	46	3	0.45625	1.046655	4.0625	1.186688	1.117422	4.3125	16.479118
2012-04-01	7	46	4	0.45625	0.54555	4.625	0.522376	0.639744	4.8125	14.131639
2012-04-01	83	73	1	0.55625	1.087845	1.0625	0.090062	1.256113	3.0625	11.6152
2012-04-01	83	73	2	0.89375	3.805826	4.6875	3.798563	3.935698	9.5	19.259487
2012-04-01	83	73	3	0.65	2.504962	4.9375	2.52	2.584484	7.4375	16.644287
2012-04-01	83	73	4	0.45625	1.715212	5.9375	1.409064	1.809402	9.5625	16.700897

For a **datapoint**, we have **4 windows** that **aggregate 64 days** historical data

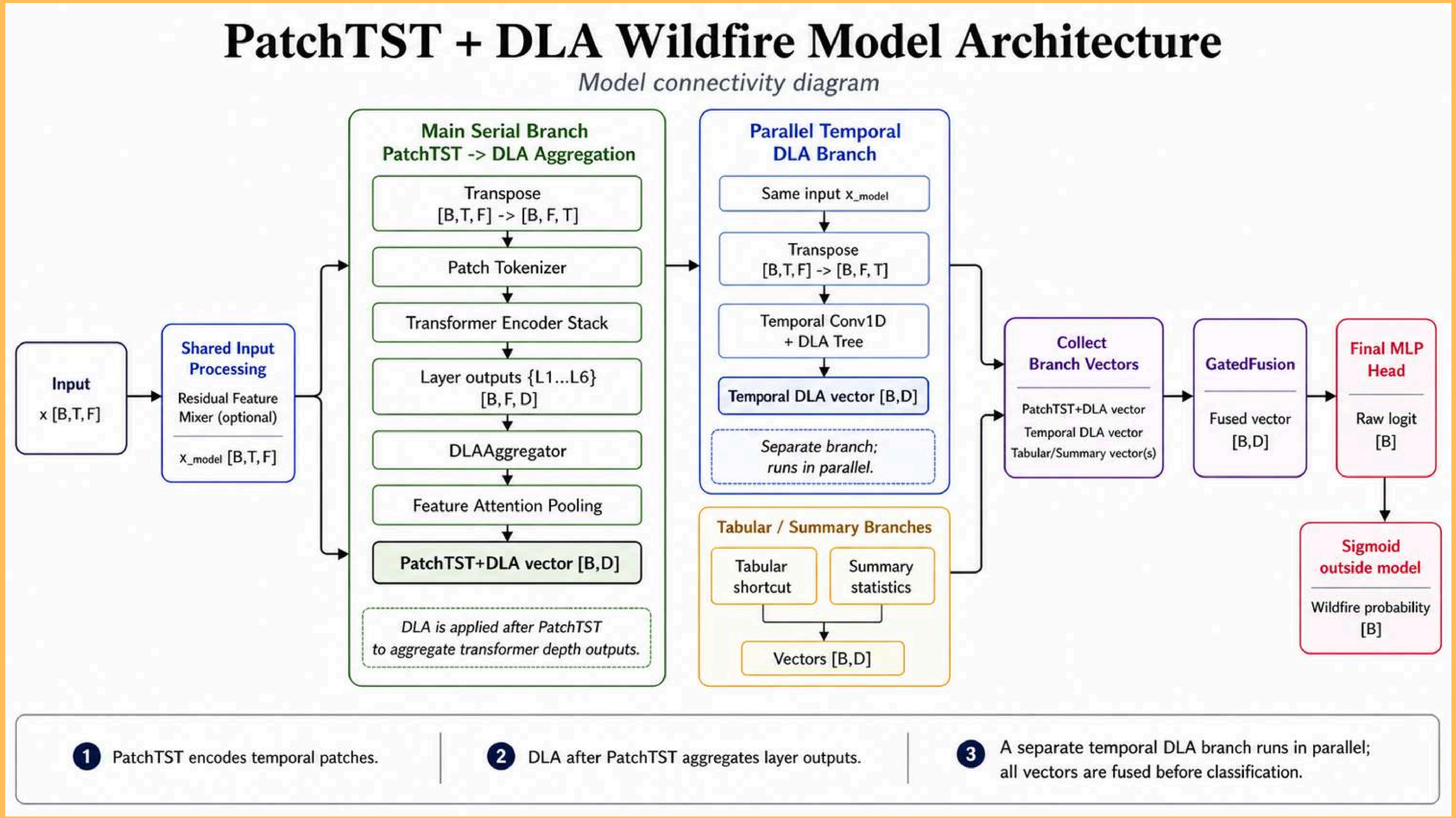
Imputed the missing data by 3x3 kernel and **rasterio** library filling method.

Here window_id = 1 represents data from (t, t-16] , window_id = 2 → (t-16, t-32] and so on....

wmWwwTs : we model What we want To see

Model Creation

Hybrid **PatchTST** Transformer + **DLA** (Deep Layer Aggregator)

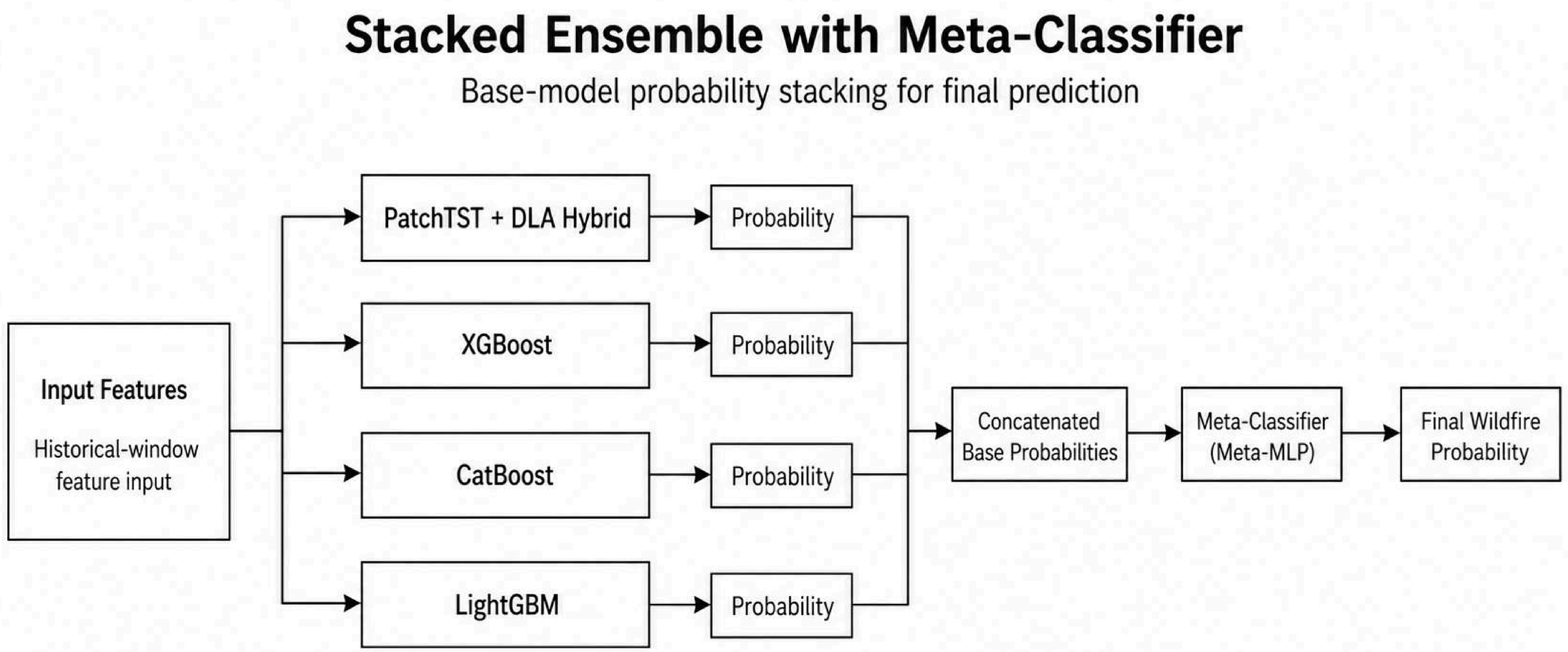


- PatchTST captures **short temporal** patterns across the 4-window history
- DLA after PatchTST combines shallow and deep transformer representations; captures **using self attention.**
- Temporal Conv1D + DLA branch captures ***local lag structure*** in a better convolutional way than self-attention.

wmWwwTs : we model What we want To see

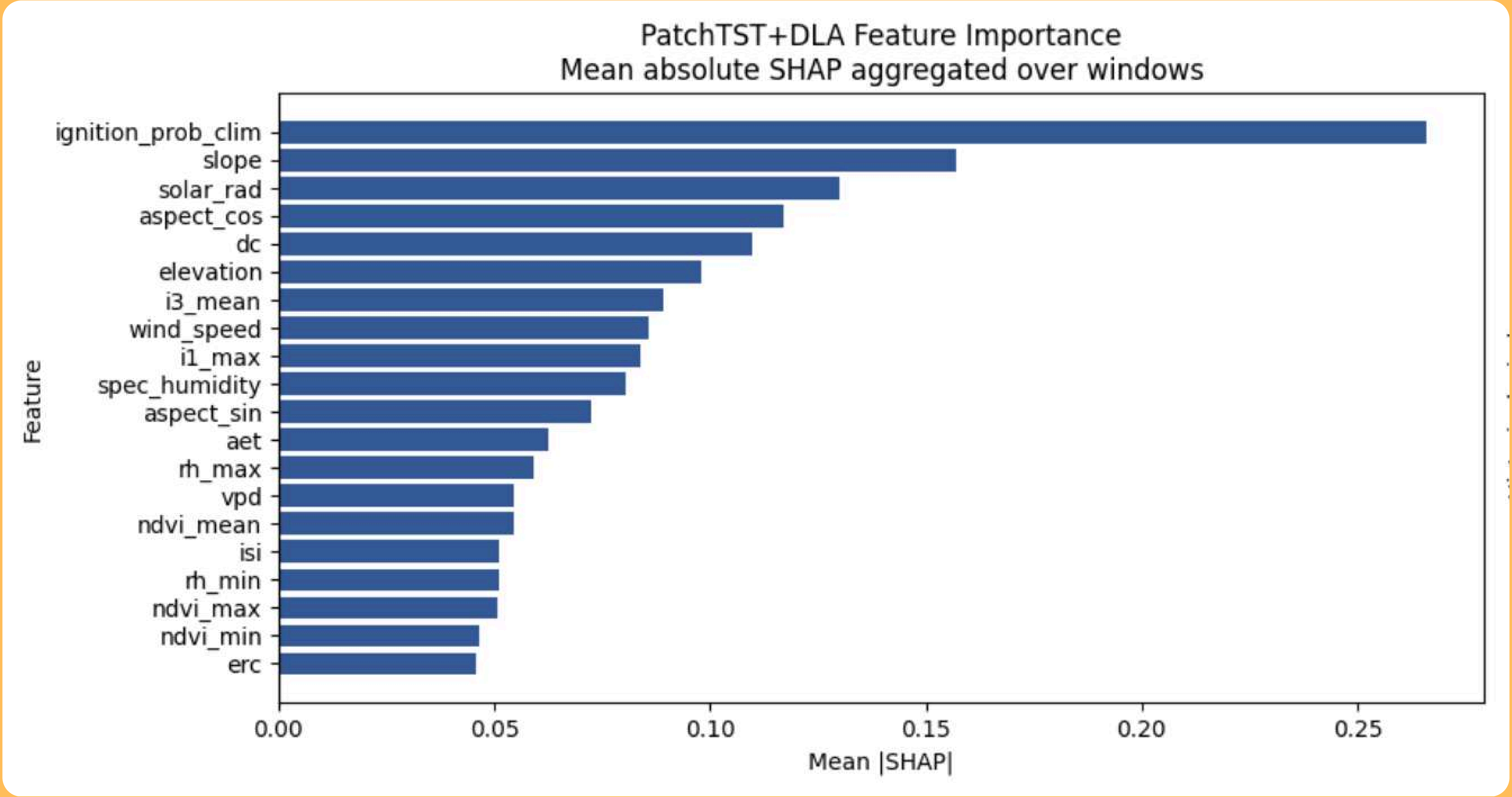
Meta Classifier

Final Model



- Tree based model included to exploit the tabular threshold behavior wmWws Vs wmWwwTs

Feature importance



wmWws Vs wmWwwTs

Analysis

Learning optimized on PR-AUC, for pipeline 1 and 2 → Best Threshold was 0.5 and 0.35

Test Metrics Comparison

Model	Precision	Recall	F1	IoU	Specificity	PR-AUC
Pipeline1 (thr=0.50)	0.0772	0.6389	0.1378	0.0740	0.9289	0.2713
Pipeline2 (thr=0.35)	0.7400	0.8222	0.7789	0.6379	0.7111	0.8765

Ablation studies

Activation

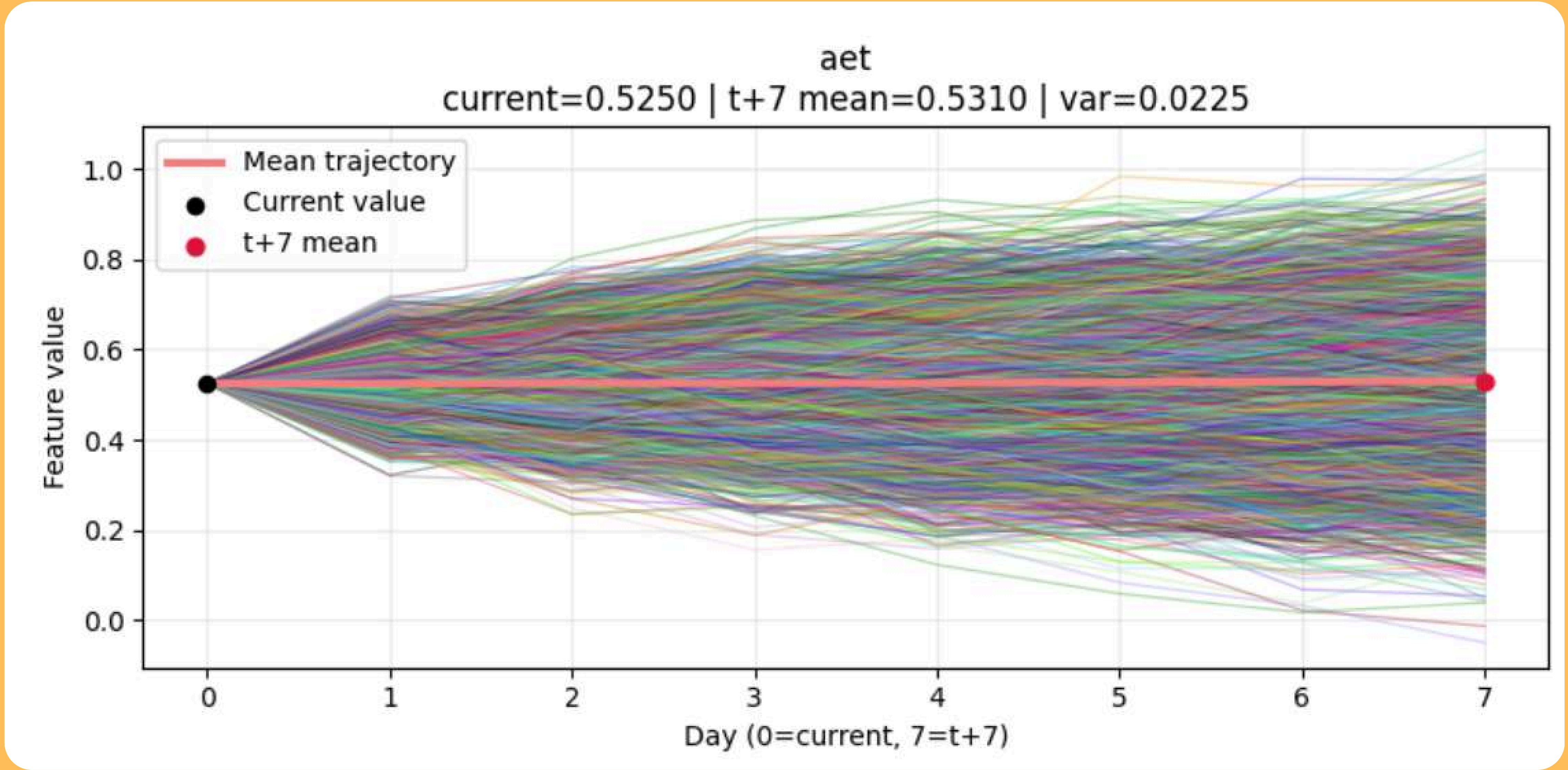
```
==== Training Meta MLP with relu ====
relu → thr=0.3500, val_PR=0.8532, test_PR=0.8765

==== Training Meta MLP with gelu ====
gelu → thr=0.3000, val_PR=0.8515, test_PR=0.8777

==== Training Meta MLP with silu ====
silu → thr=0.3000, val_PR=0.8503, test_PR=0.8774

==== Training Meta MLP with leaky_relu ====
leaky_relu → thr=0.3500, val_PR=0.8523, test_PR=0.8756
```

Future Forecasting by path simulations

Monte Carlo

Future **Probability** will depend on the **future feature values**

For a datapoint, loop along features and also along the historical 64 days.
Calculate **mean** and **variance** for each of the feature from the historical data



$$x_{t+k}^{(s)} = x_{t+k-1}^{(s)} + \epsilon_{t+k}^{(s)}, \qquad k = 1, \ldots, 7$$

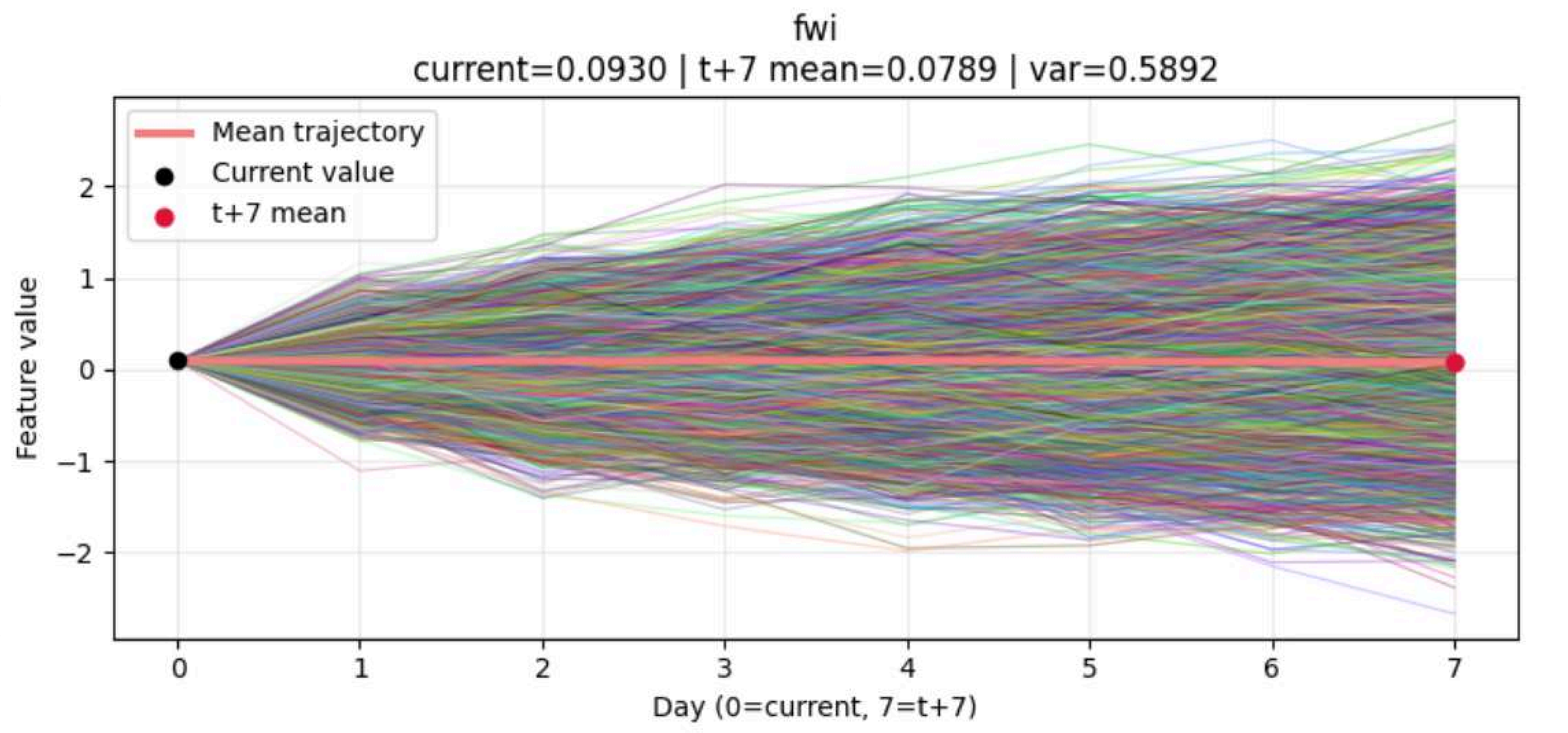
$$\epsilon_{t+k}^{(s)} \sim \mathcal{N}(0, \sigma^2)$$

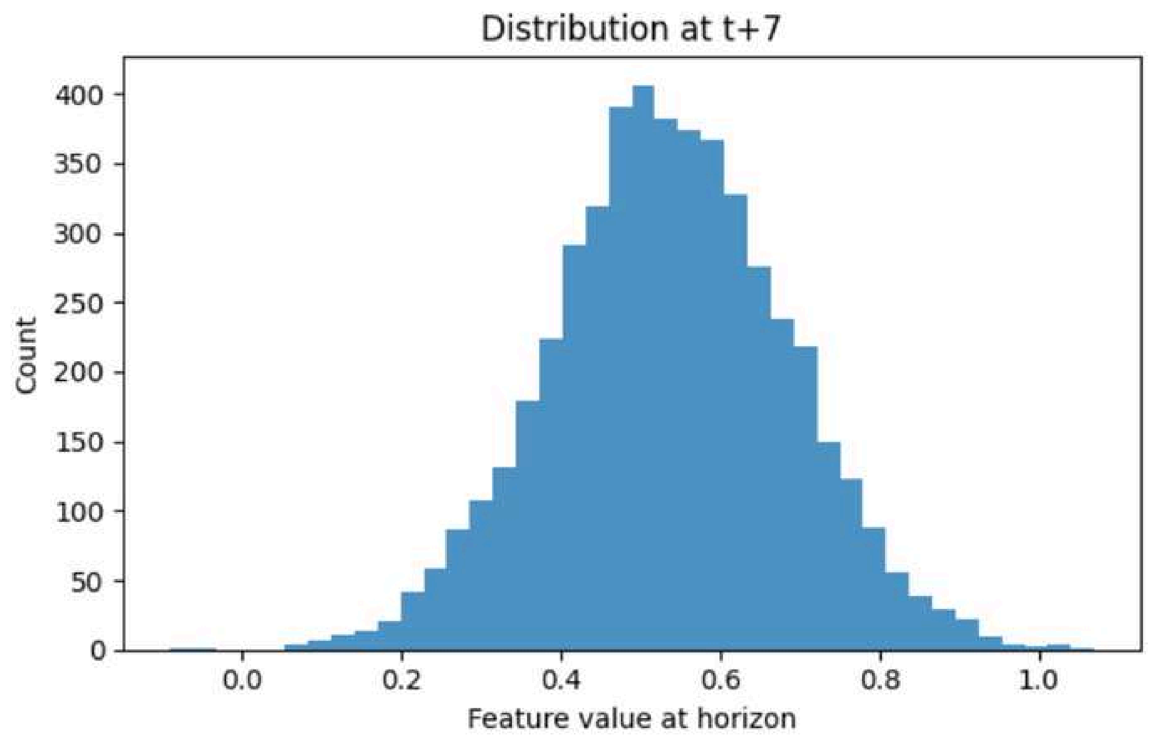
Using the mean projected value for training the MLP layer

Future Forecasting

Monte Carlo Stats

Generate **5000** sample for each feature, for each **projected** values train a MLP model
The model will output a probability for each sample, the **mean** of all the sample probability would be the **future forecasted** wildfire probability.





Future Forecasting

Monte Carlo Stats

- Starting from current day values, FWI and AET are projected to t+7 (mean) using Monte Carlo simulation
- Multiple simulations are run to capture uncertainty, Mean and standard deviation at t+7 are obtained from the simulations
- The **MLP layer** is then used to estimate fire probability at t+7 (**forecast_fire_probability**)

```
================================================================================
[Timing Summary]
Historical stats time        : 0.000 seconds
Deterministic forecast time : 0.002 seconds
Monte Carlo forecast time    : 0.035 seconds
Total forecast runtime       : 0.058 seconds
Total MC samples             : 5000
Average time per MC sample   : 0.000007 seconds
================================================================================
Feature summary:
```

	feature	current_day_value	t_plus_7_mc_mean_value	t_plus_7_mc_variance	t_plus_7_mc_std
0	aet	0.525000	0.531007	0.022452	0.149841
1	fwi	0.092951	0.078923	0.589200	0.767594

```
Fire probability summary:
```

	target_date	row	col	initial_fire_probability_at_t	forecast_fire_probability_t_plus_7_mean	forecast_fire_probability_t_plus_7_std	deterministic_fire_probability_t_plus_7
0	2012-04-01	7	46	0.083333	0.412306	0.000211	0.412306

Hugging Face Datasets (Custom Dataset)

https://huggingface.co/datasets/NagrajMG/WildFire-X

https://huggingface.co/datasets/NagrajMG/WildFire-Y

Kaggle Source Code (Custom Model)

For future + 7 forecasting:
https://www.kaggle.com/models/nagrajgaonkar/cv5600-submission-monte-carlo

For current day prediction:
https://www.kaggle.com/models/nagrajgaonkar/nagrajmg